RELATIVE PSEUDO-COHERENCE IS PRESERVED BY TOR-INDEPENDENT BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The two derived tensor products in the base change diagram agree, so K tensor_A^L A' = K tensor_R^L R' = K tensor_{R[x]}^L R'[x]. invoked at proof lines 21558-21565. Identifies the base-changed complex over the polynomial presentation.
- Derived tensor product along the ring map R[x] -> R'[x] preserves relative pseudo-coherence. invoked at proof lines 21565-21566. Concludes the R'-relative pseudo-coherence.

Source reference. Original lemma 0661: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14955–14955. Cited at proof lines 21564-21565 (applied twice) to identify the derived base change; the conclusion is used inside the counted span.

Source reference. Original lemma 0650: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16345–16345. Cited at proof lines 21565-21566 to conclude the R'-relative pseudo-coherence; the conclusion is used inside the counted span.

Lemma 0.2. *Let $R \to A$ be a finite type ring map. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a complex of A-modules which is m-pseudo-coherent (resp. pseudo-coherent) relative to R. Let $R \to R'$ be a ring map such that A and R' are Tor independent over R. Set $A' = A \otimes_R R'$. Then $K^\bullet \otimes^{\mathbf{L}}_A A'$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R'.*

Proof. Choose a surjection $R[x_1, \ldots, x_n] \to A$. Note that

$$K^\bullet \otimes^{\mathbf{L}}_A A' = K^\bullet \otimes^{\mathbf{L}}_R R' = K^\bullet \otimes^{\mathbf{L}}_{R[x_1,\ldots,x_n]} R'[x_1, \ldots, x_n]$$

by Lemma 0661 applied twice. Hence we win by Lemma 0650. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.